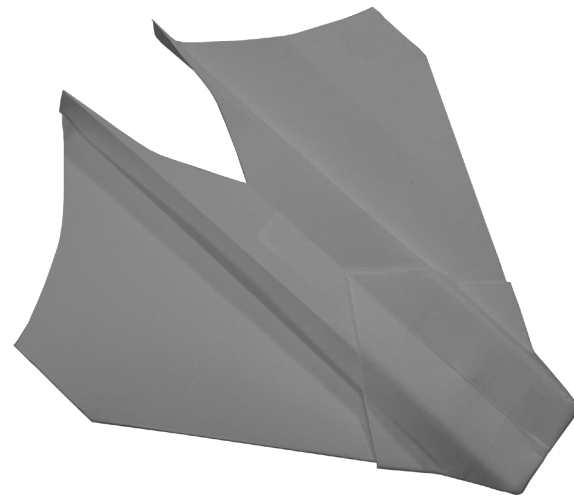


P15 MANTA



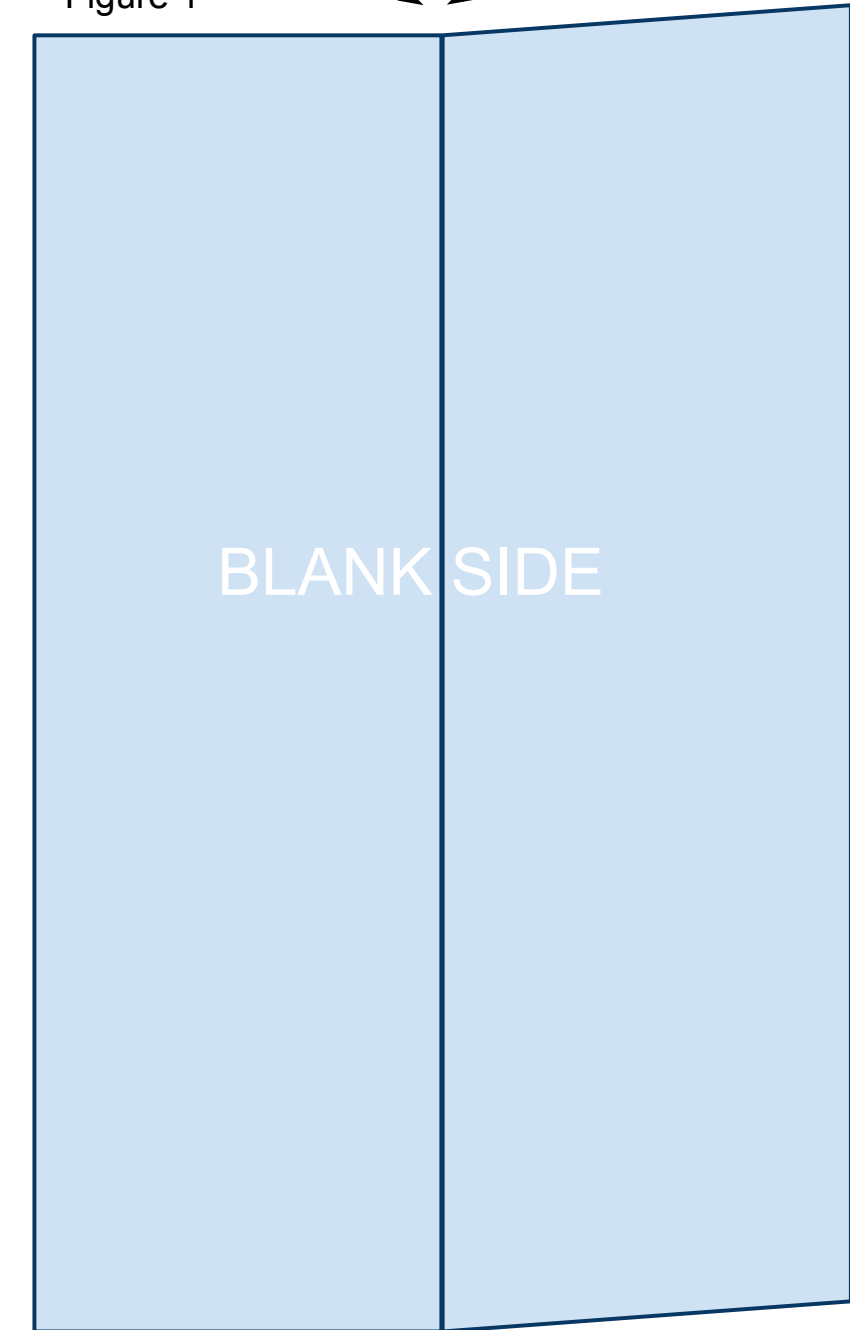
The printable template linked below is 4.25in x 5.5in. That is equivalent to half of an 8.5in x 11in sheet of printer paper. While the designs in this book will fly great with regular printer paper, I recommend 32lb/ream paper. The stronger paper will hold its shape better during flight allowing for higher and more exciting flights. Once you become comfortable making the planes from this template you can simply cut regular printer paper in half in order to start. Print the PDF template located at the bottom of the web page: <https://sites.google.com/site/afspaperairplanes/p15>

In case your reading device is not connected to the internet, doing a Google search for “AFS Paper Airplanes” should get you to the webpage.

1

Cut out the template and place the paper on the table so that the printed information is on the underside with the printed triangle facing away from you. You will be looking at the blank side. Bring the right side up and fold the paper in half the long way. Unfold leaving a visible crease as shown in Figure 1 below.

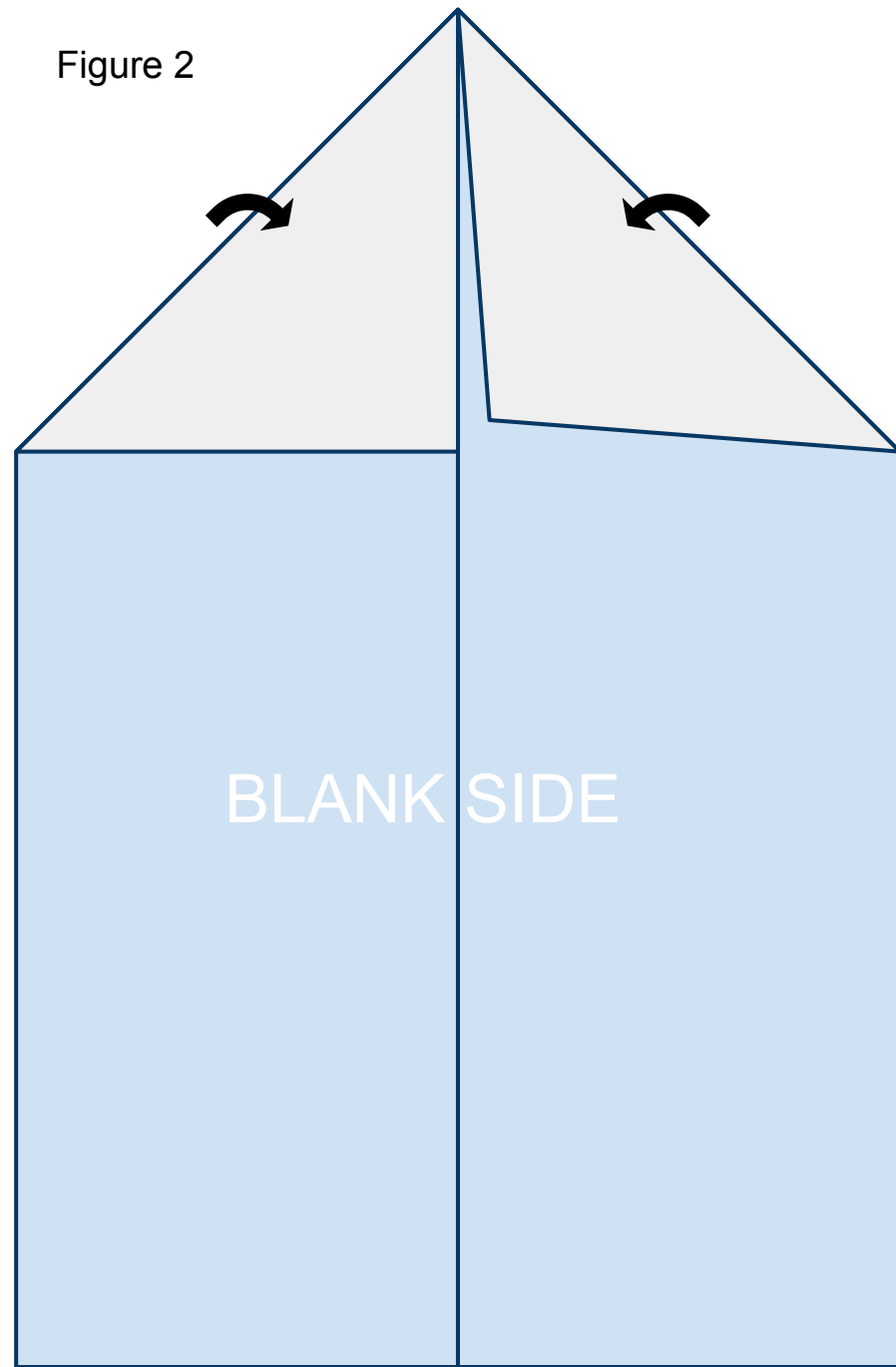
Figure 1



2

Fold the leading corners back along the printed solid lines. The edge of the paper should line up along the crease you created in step 1 as shown in Figure 2 below.

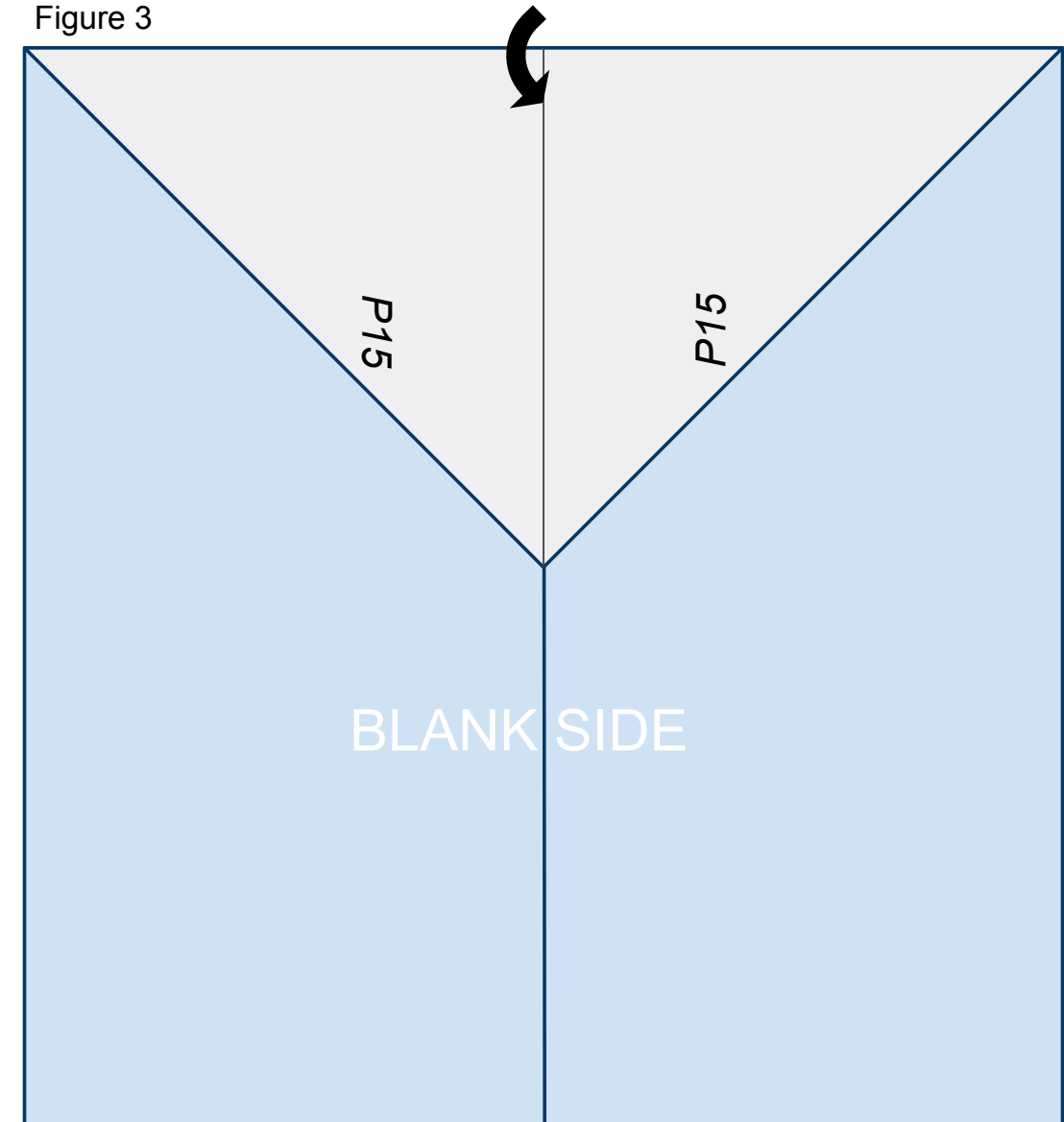
Figure 2



3

Fold the nose of the plane created in step 2 back along the solid line printed on the front of the paper so that it lines up on the center crease created in step 1 as shown in Figure 3 below.

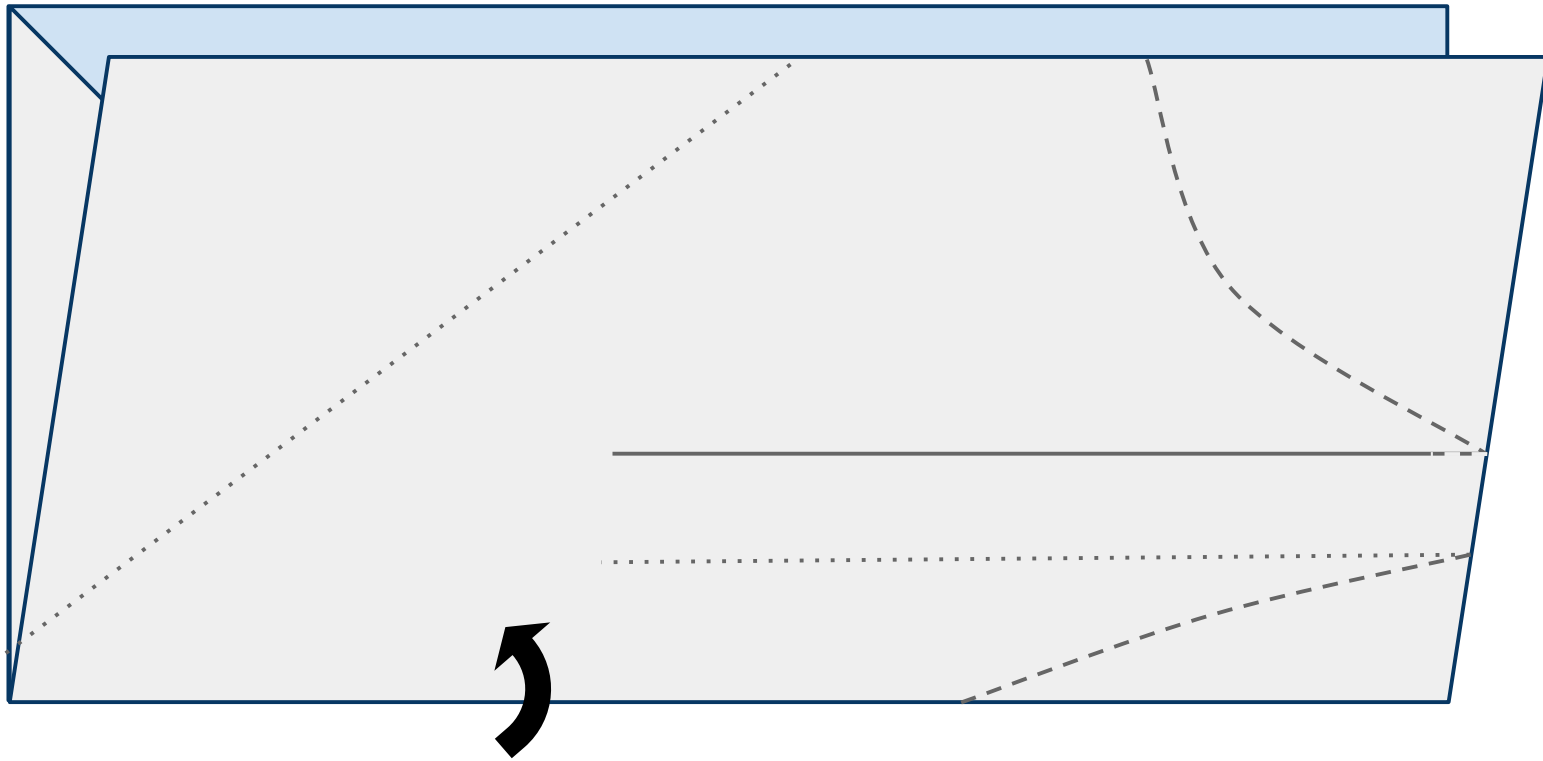
Figure 3



4

Turn the paper to the left (counterclockwise) and fold the paper upward in half hiding the triangle created in the previous steps refolding along the fold from step 1 as shown in Figure 4 below.

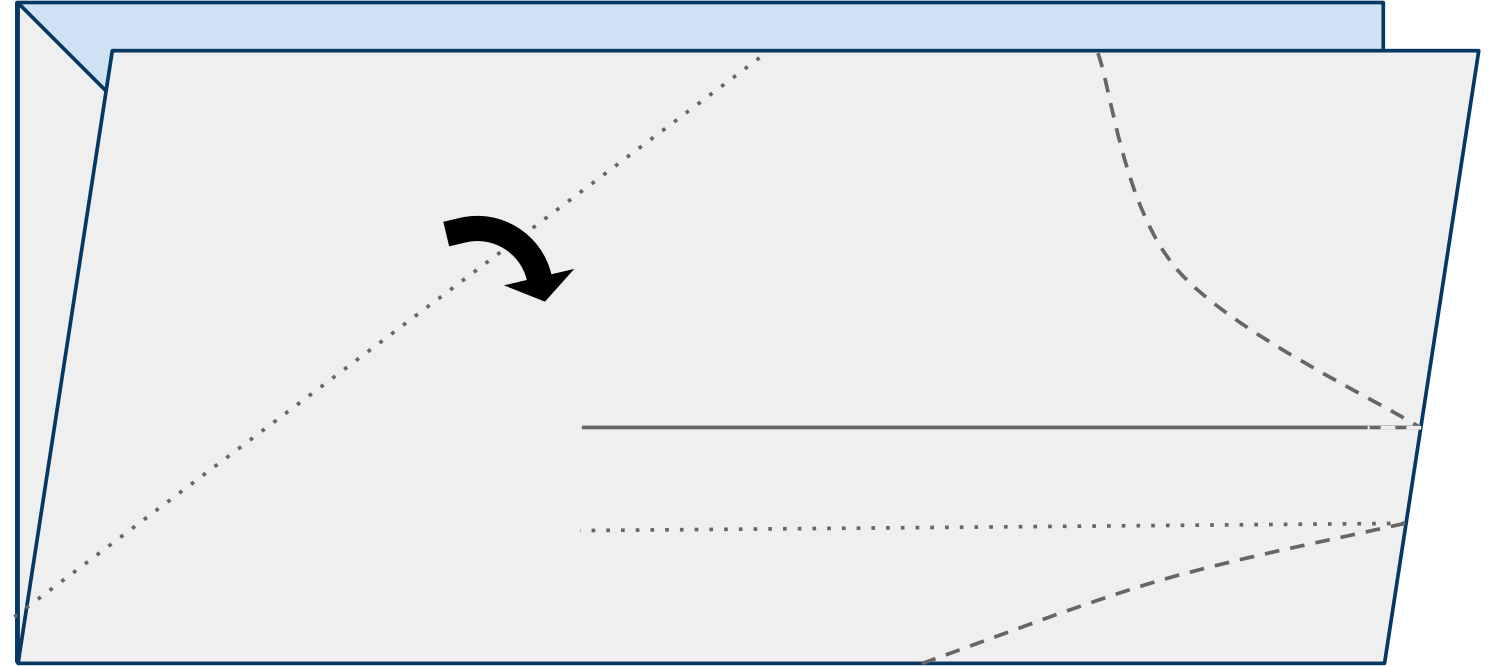
Figure 4



5

Fold the leading edge of the near side of the plane down along the dotted line as shown in Figure 5 below.

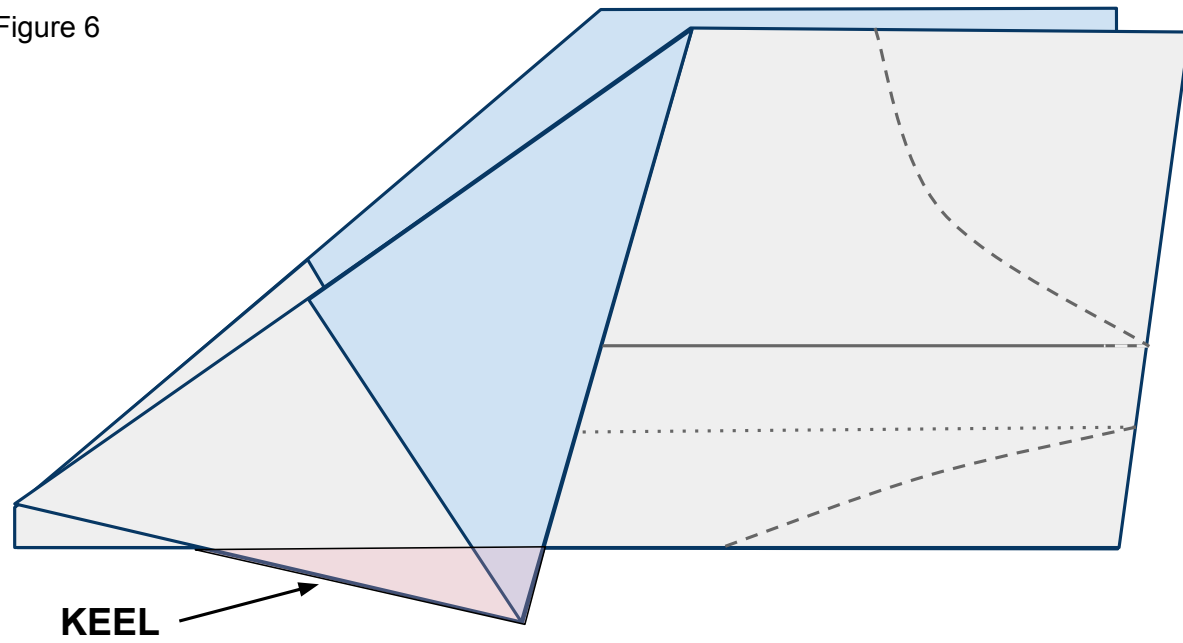
Figure 5



6

Do the same to the other side of the plane. It is important to match these 2 folds exactly so that your plane is symmetrical and will fly straight. The area that is created below the body and designated in pink will be called the keel.

Figure 6



7

Place tape along the keel as shown in Figure 7 below and cut a slit in the tape along the red line. Now fold the tape around and under both sides of the plane and keel securing them together. See Figure 7b for what the far side of the plane looks like.

Figure 7b

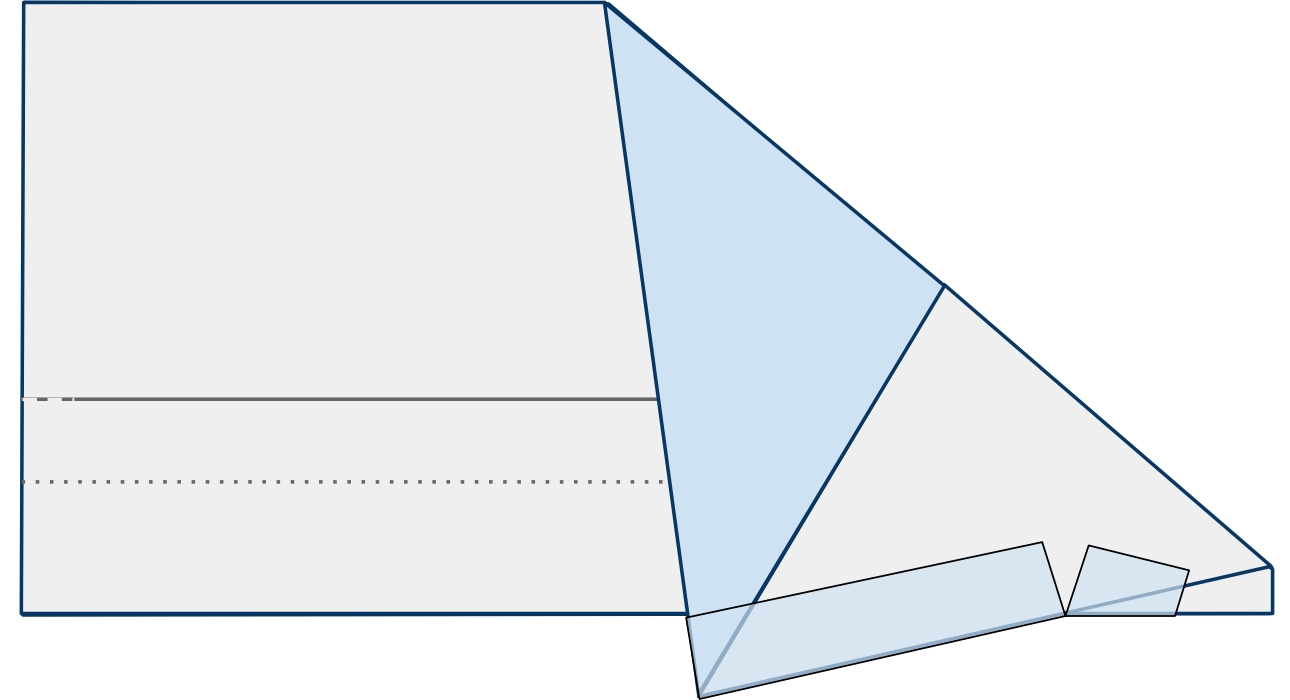
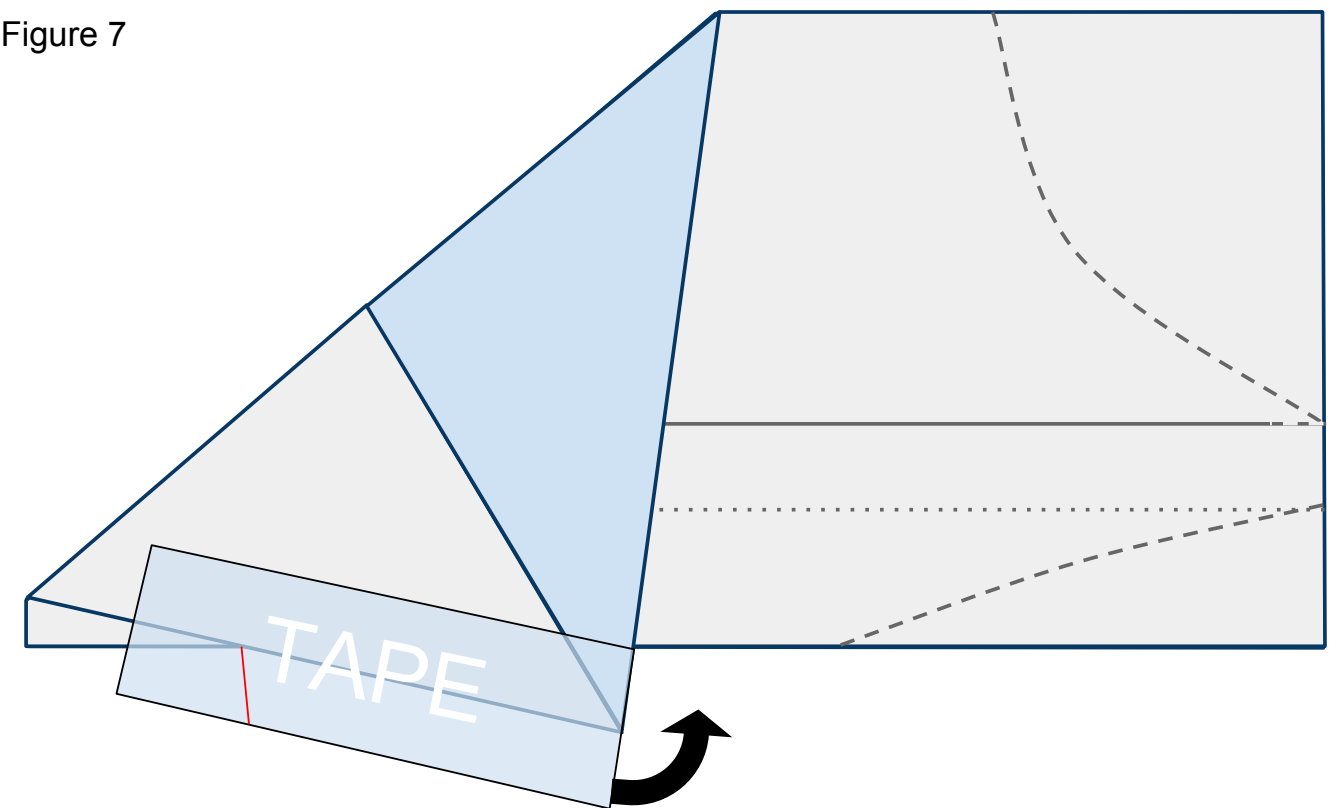


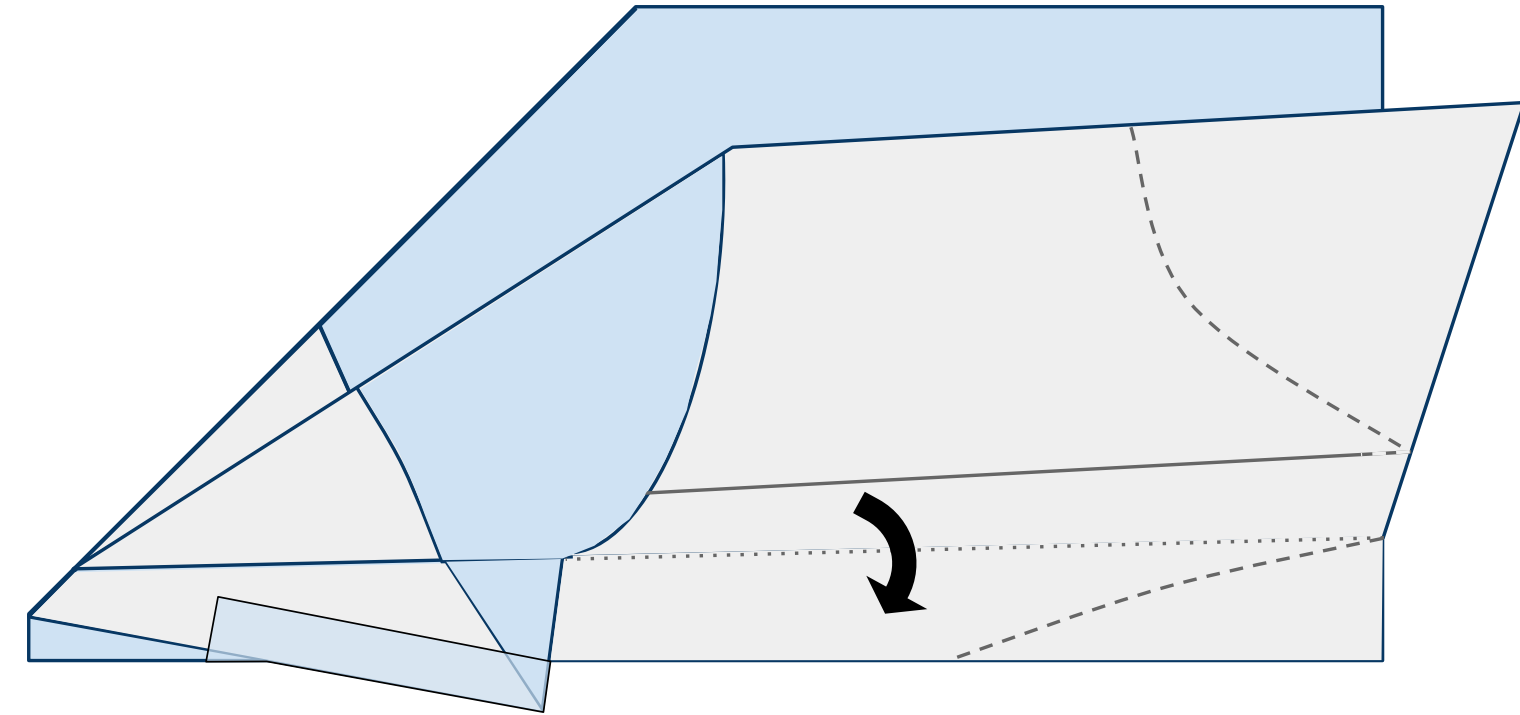
Figure 7



8

Fold down the near side wing along the dotted line as shown in Figure 8 below.

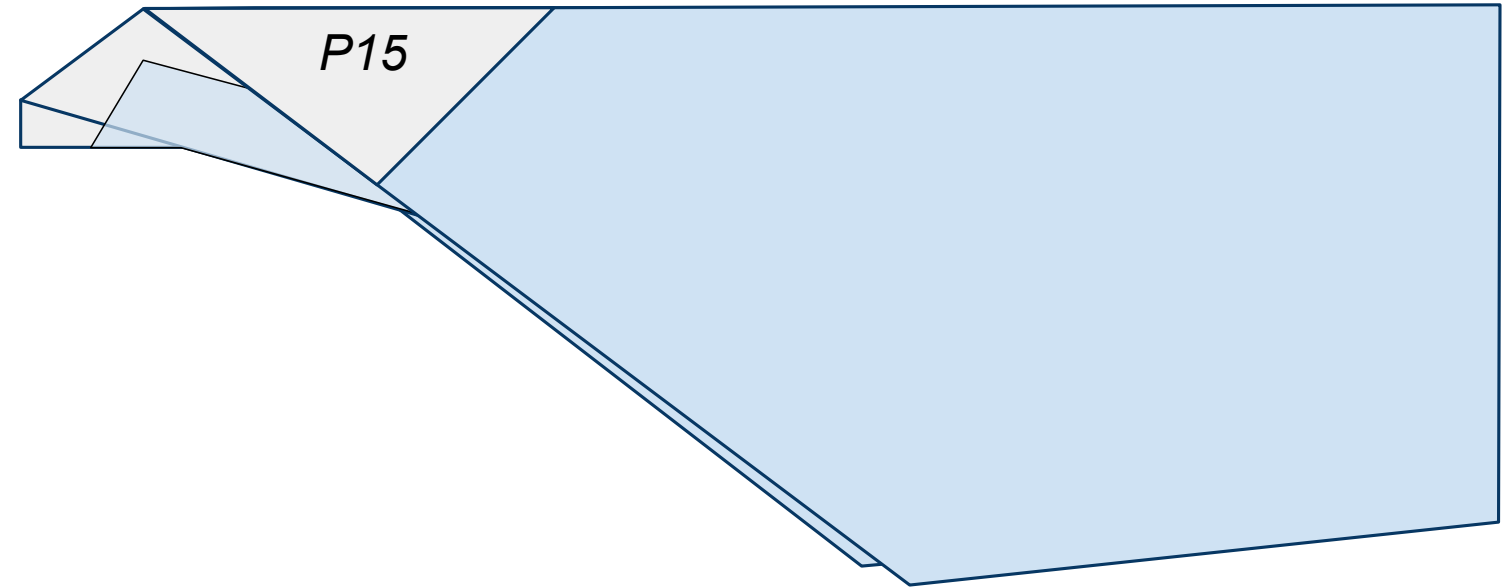
Figure 8



9

Fold down the far side wing along the dotted line also as shown in Figure 9 below. Insure that these folds are as symmetrical as possible. If you need to choose, always choose symmetry over following the lines printed on the paper.

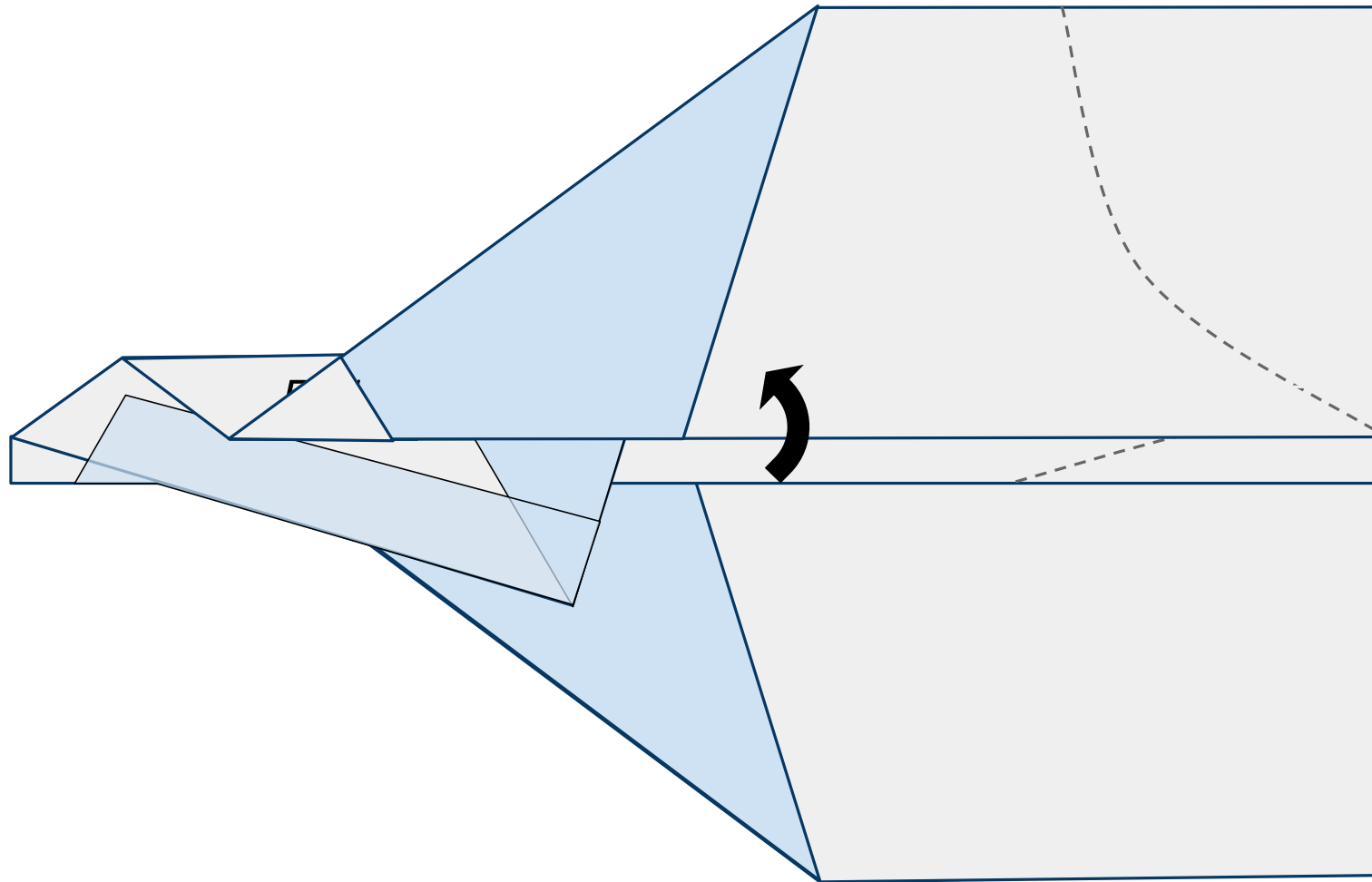
Figure 9



10

Fold the near side wing up along the printed solid line as shown in Figure 10 below.

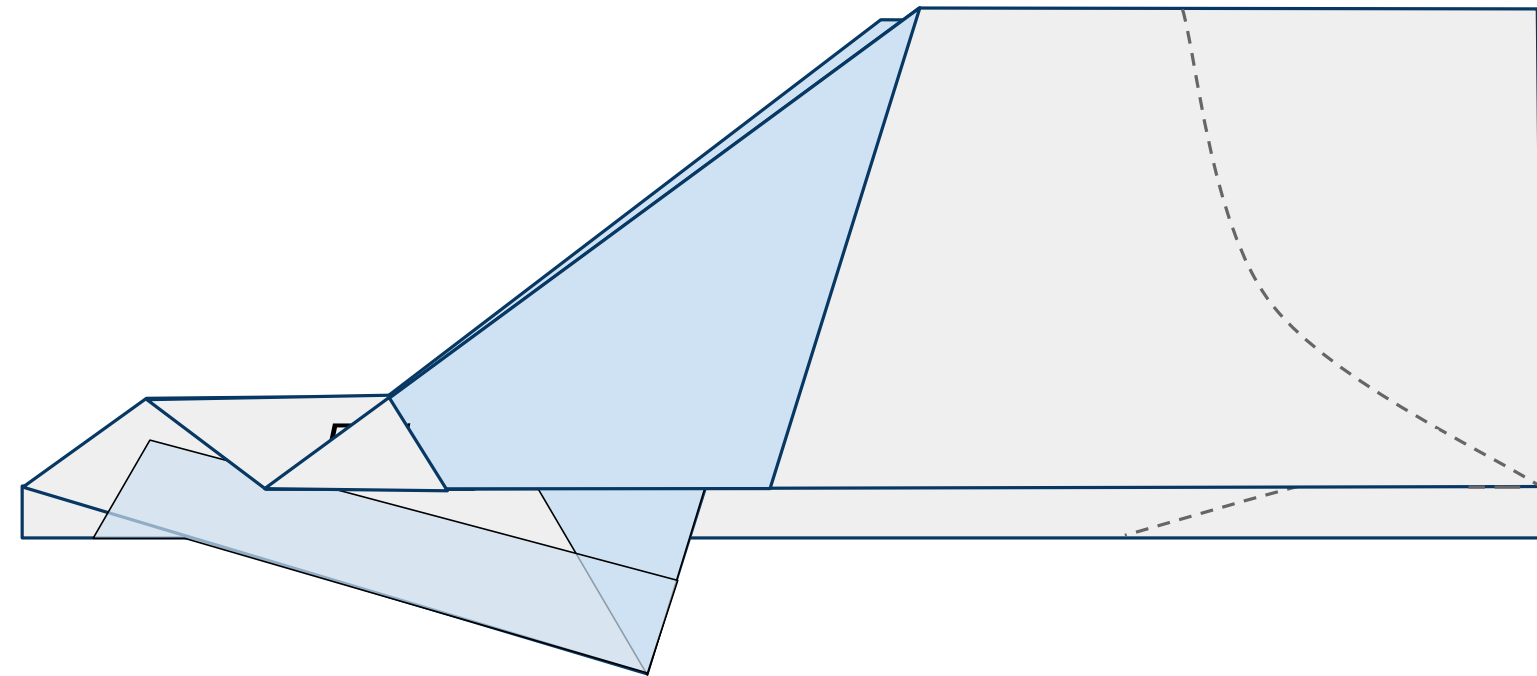
Figure 10



11

Do the same to the other side. Fold the far side wing up along the sold line on the other side of the plane as shown below in Figure 11 below.

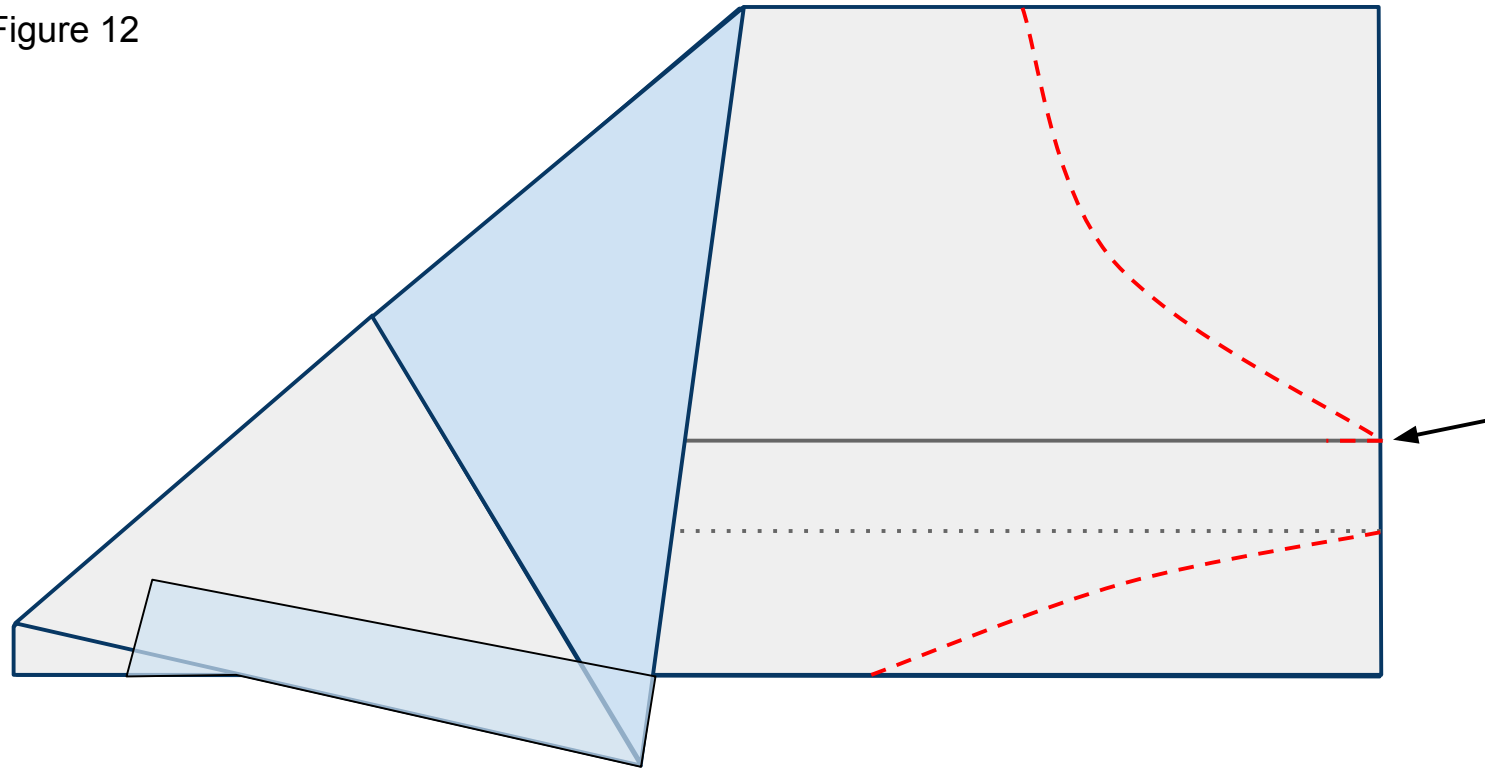
Figure 11



12

Unfold the wings that were folded in steps 8-11 and lay the plane flat on your surface as shown in Figure 12 below. Cut through both layers of paper along the red dashed lines. Be sure to include the short horizontal cut line along the wing's fold. This cut will help create the plane's flaps.

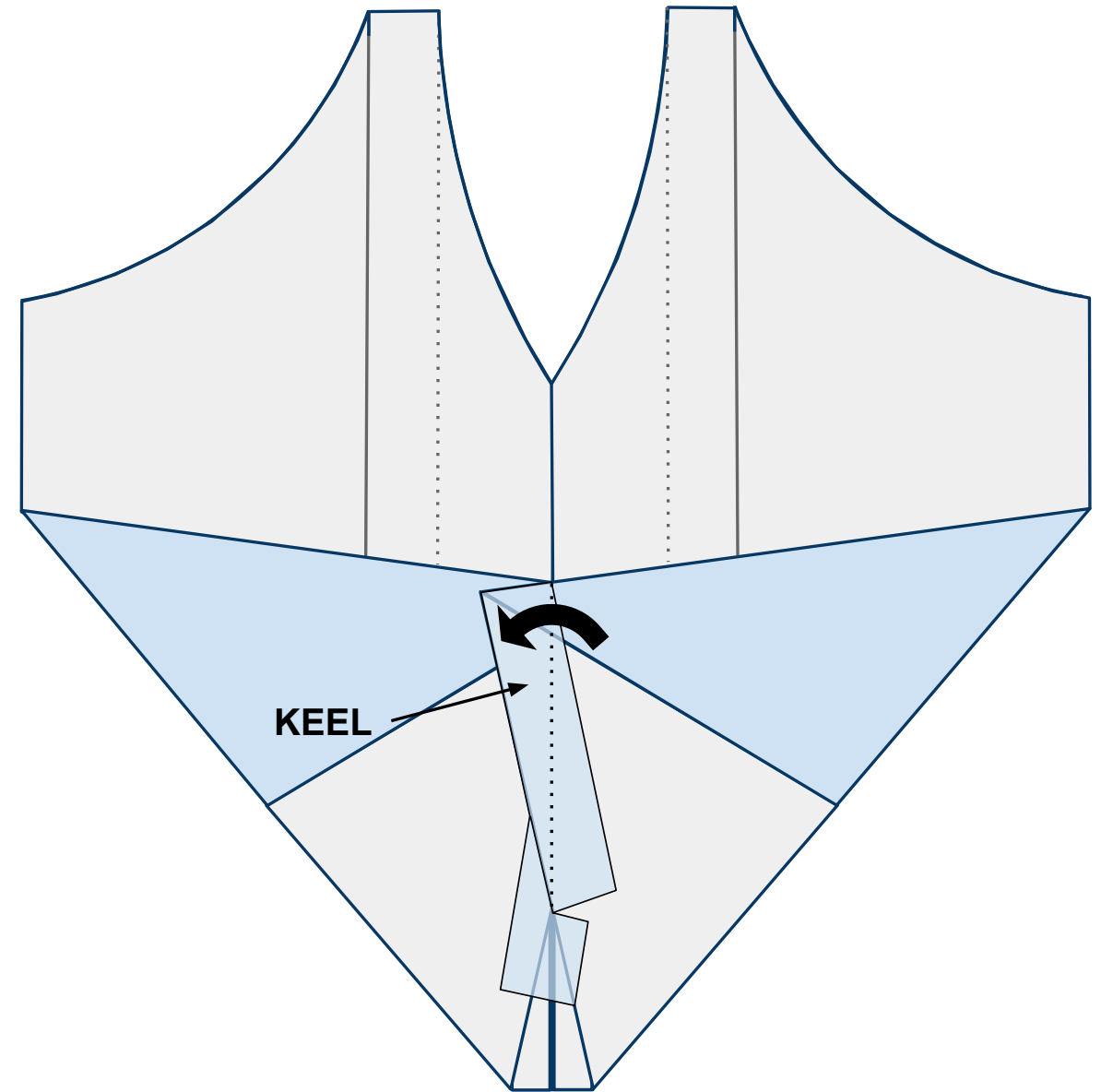
Figure 12



13

Open the plane up flat on your work surface with the nose toward you and looking at the underside of the body. When you do this the keel will want to stand up off the table. Bend it to the left along the center line and give it a strong crease as shown in Figure 13 below.

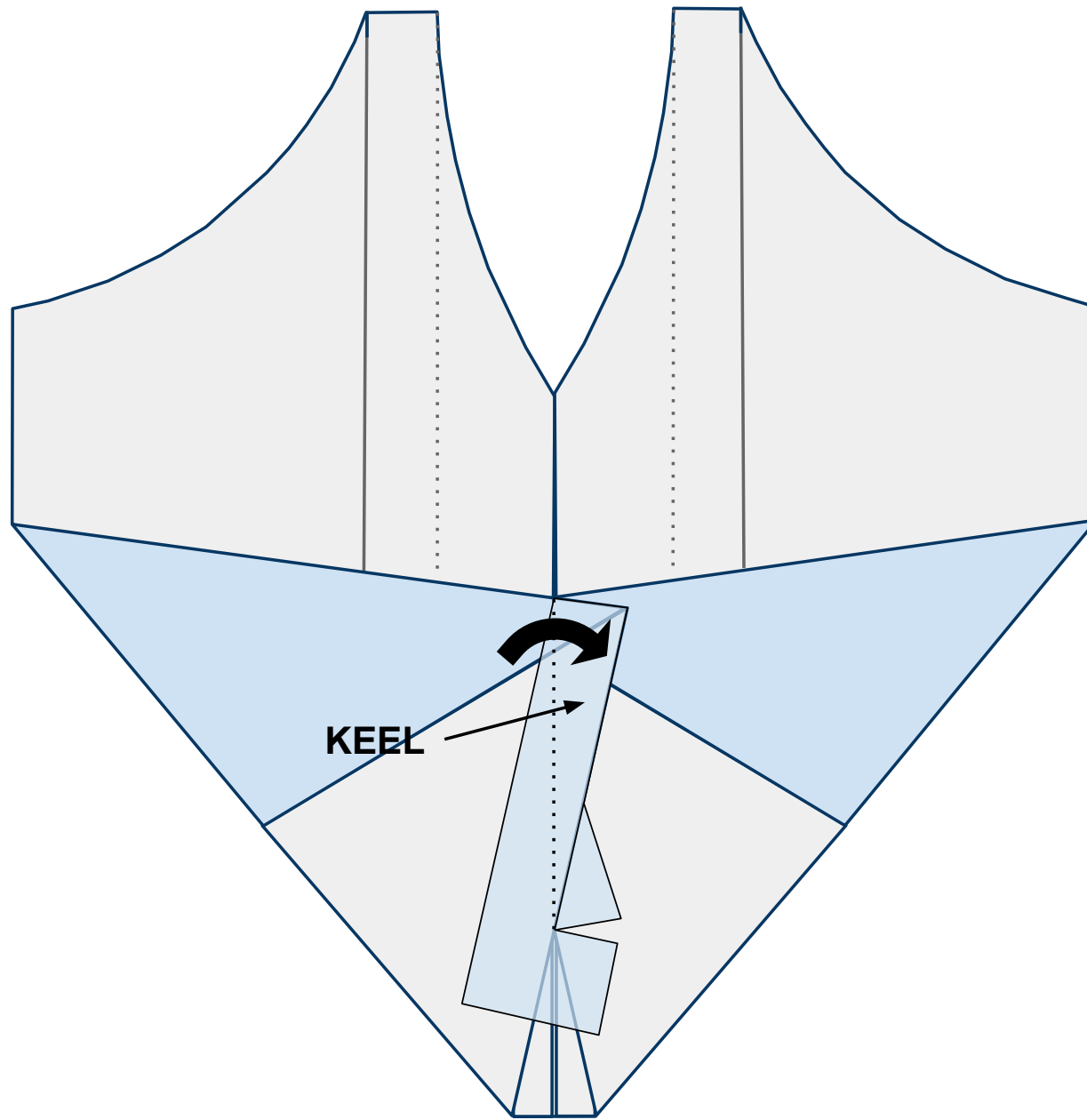
Figure 13



14

Now fold the keel to the right side creating a strong crease as shown in Figure 14 below. After making this crease, allow the keel to remain upright for the remainder of construction.

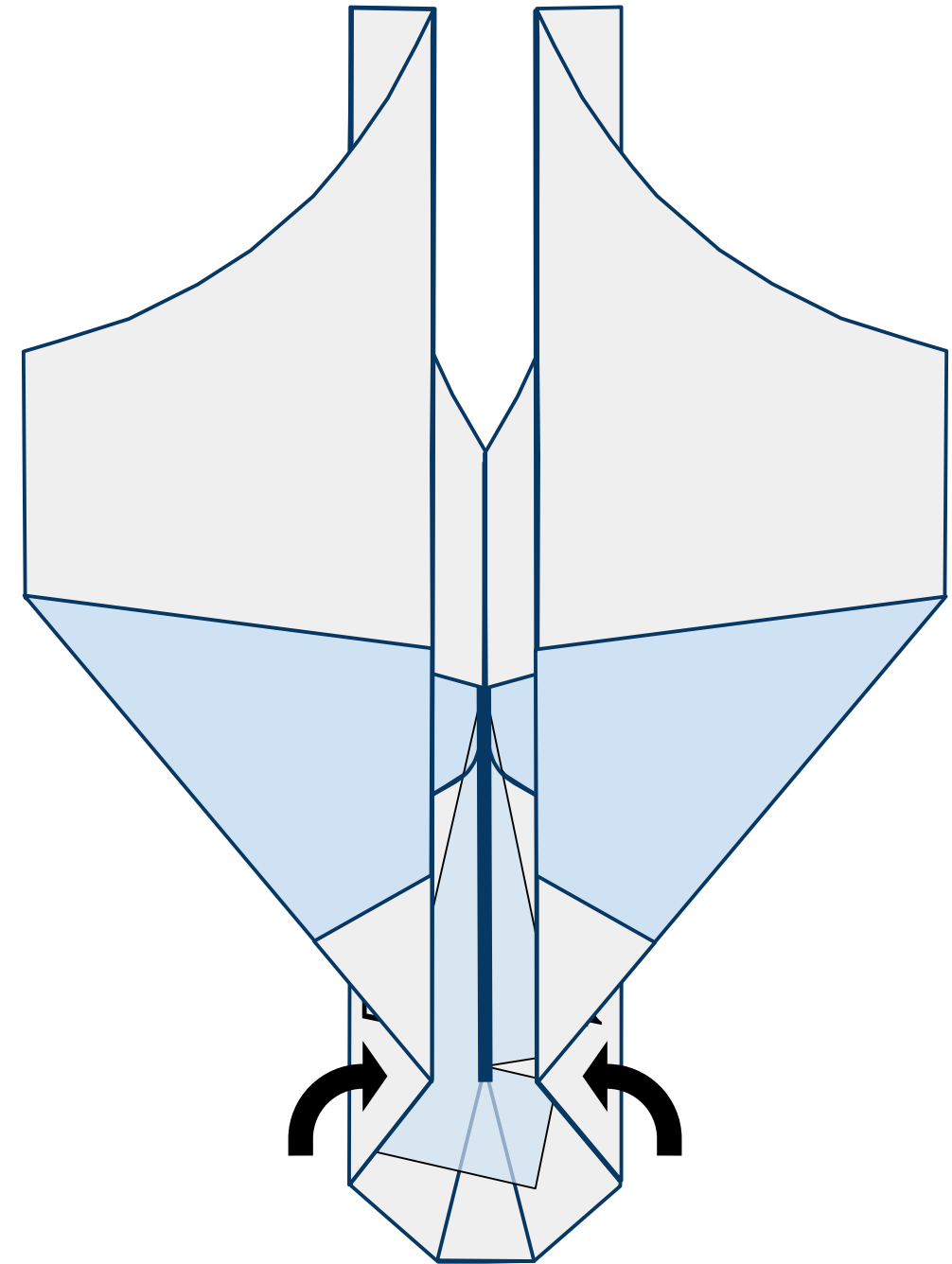
Figure 14



15

The keel is now represented by the thick **blue** line. Allowing the keel to remain vertical, bring the wings back in along their creases created in steps 8-11 as shown in Figure 15 below.

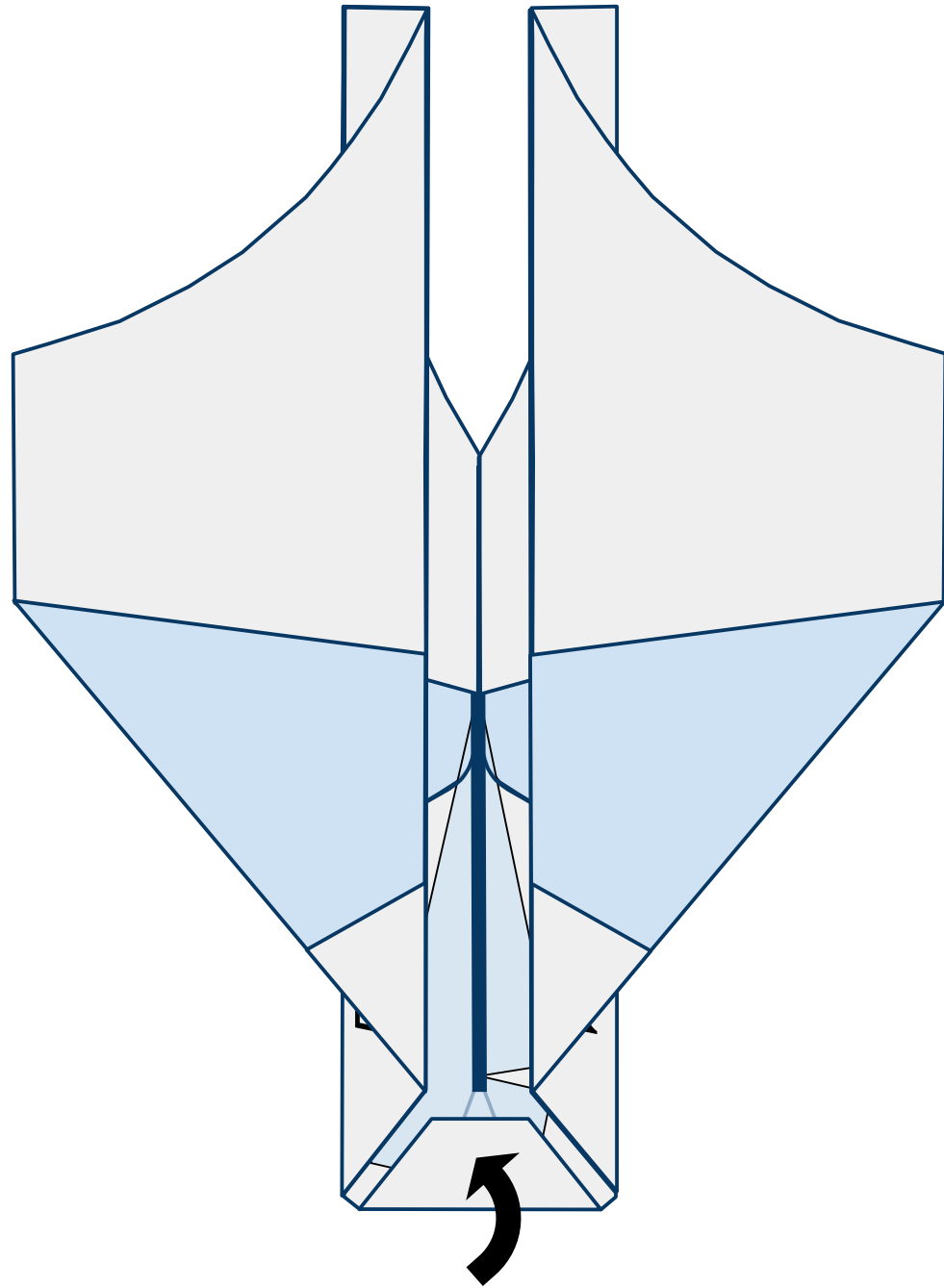
Figure 15



16

Fold the nose of the plane back and in as shown in Figure 16 below.

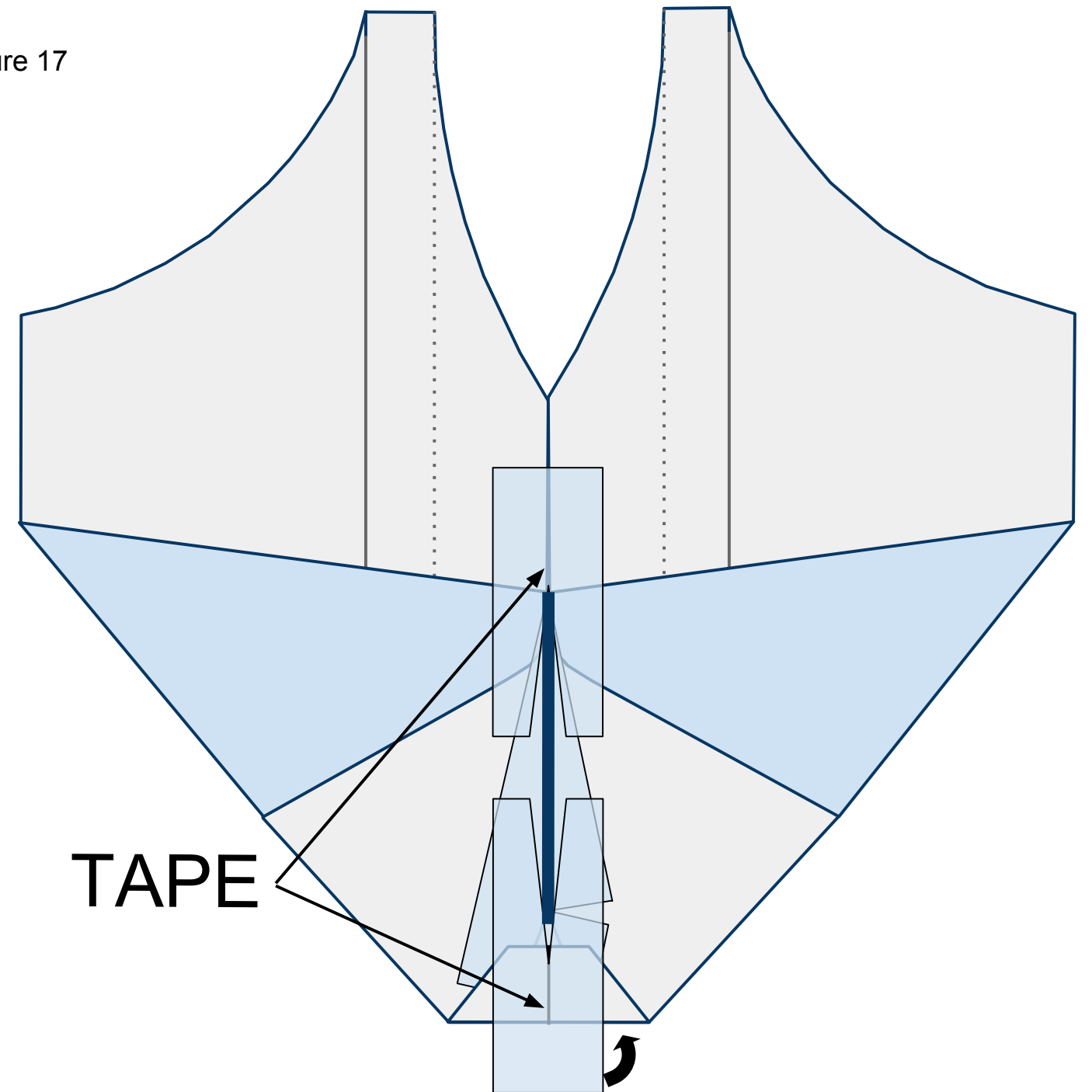
Figure 16



17

Push the wings back out and flat on the table. Place 2 pieces of tape with slices cut in them on the exposed underside of the plane in front of and behind the keel as shown in Figure 17 below. Wrap the tape around the nose to the top side of the plane.

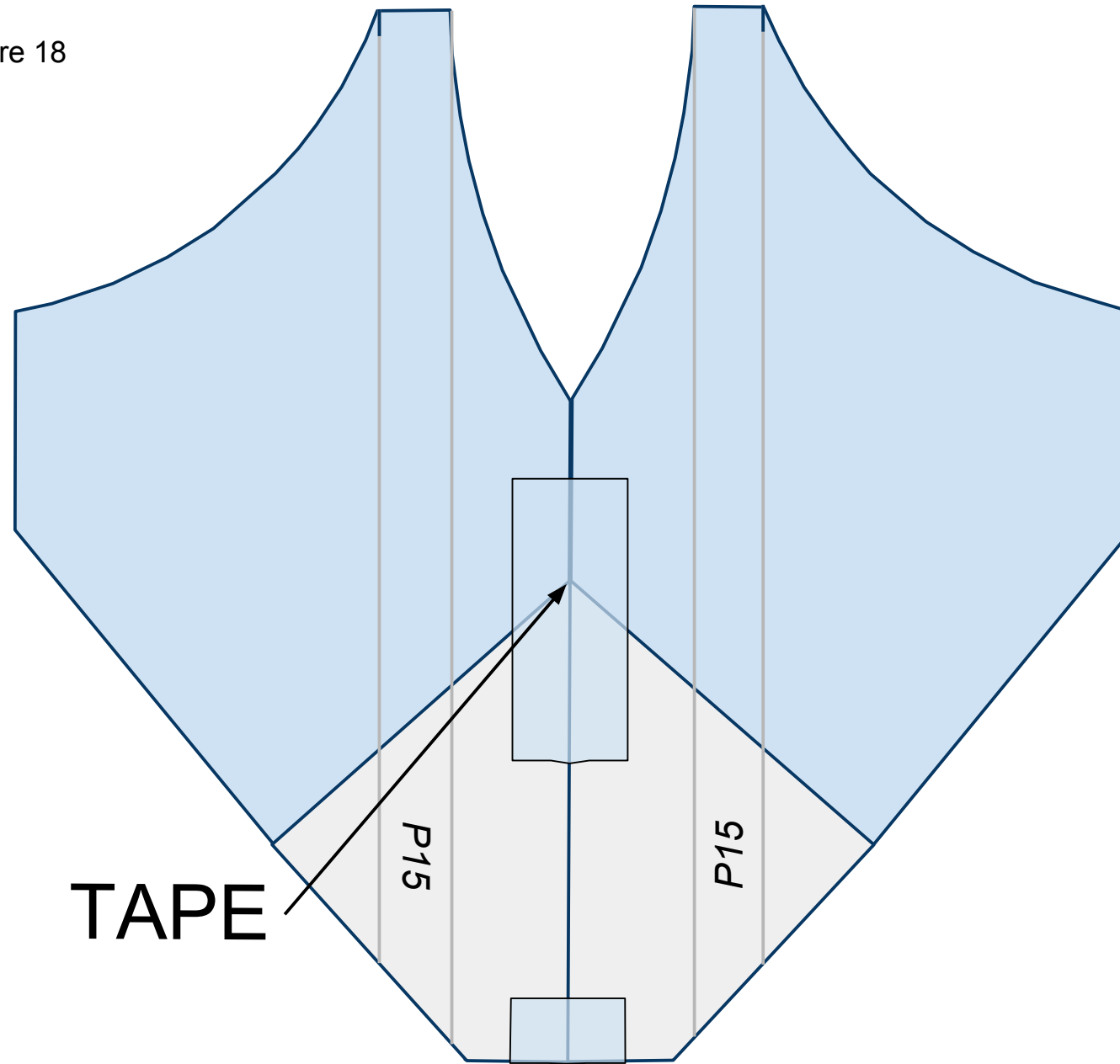
Figure 17



18

Turn the plane over laying the keel to one side. Place tape along the fuselage as shown in Figure 18 below.

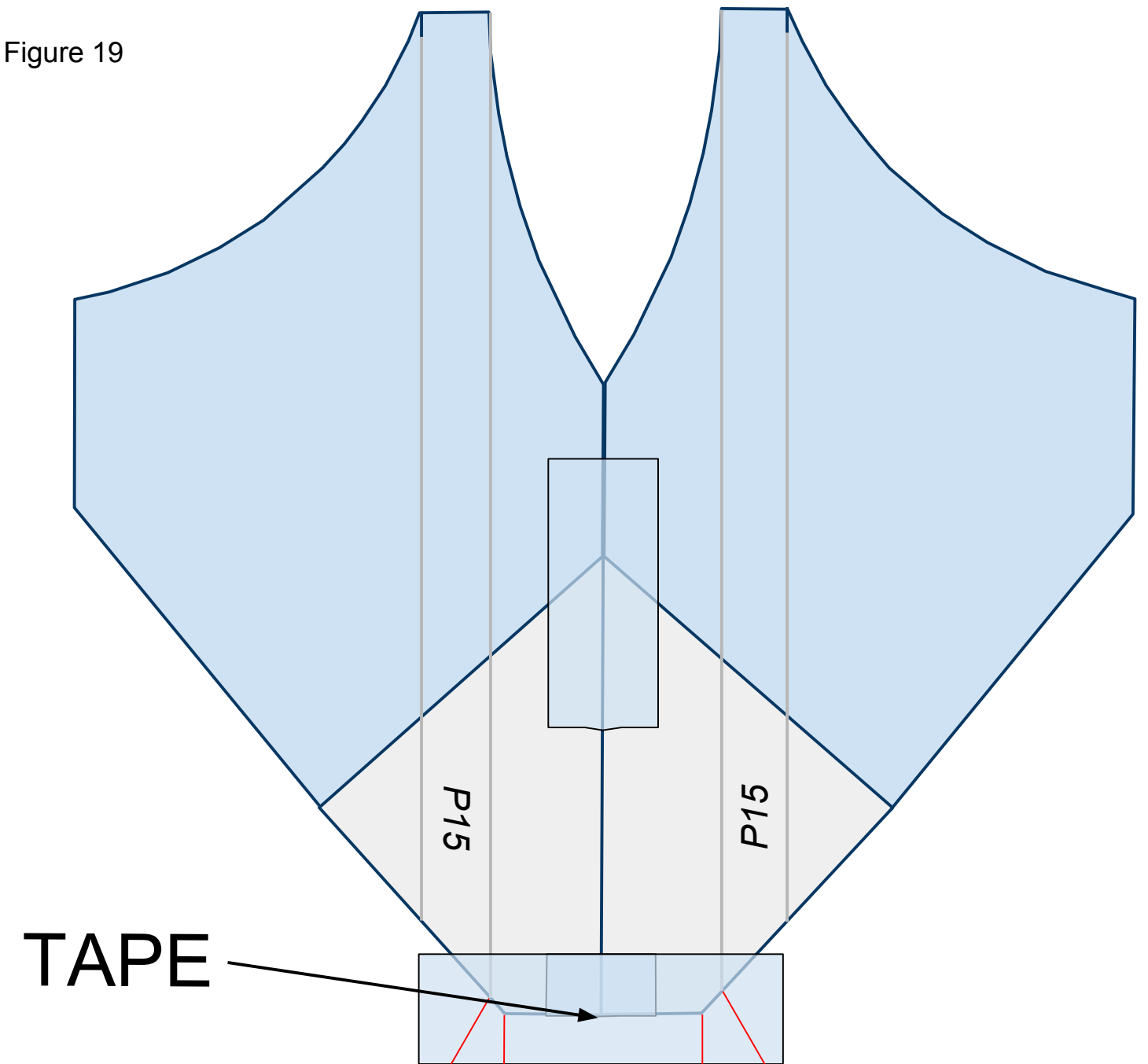
Figure 18



19

Place a piece of tape across the nose of the plane as shown in Figure 19 below. Cut slits along the **red** lines as shown. These slits will help you fold the tape under the plane in the next step.

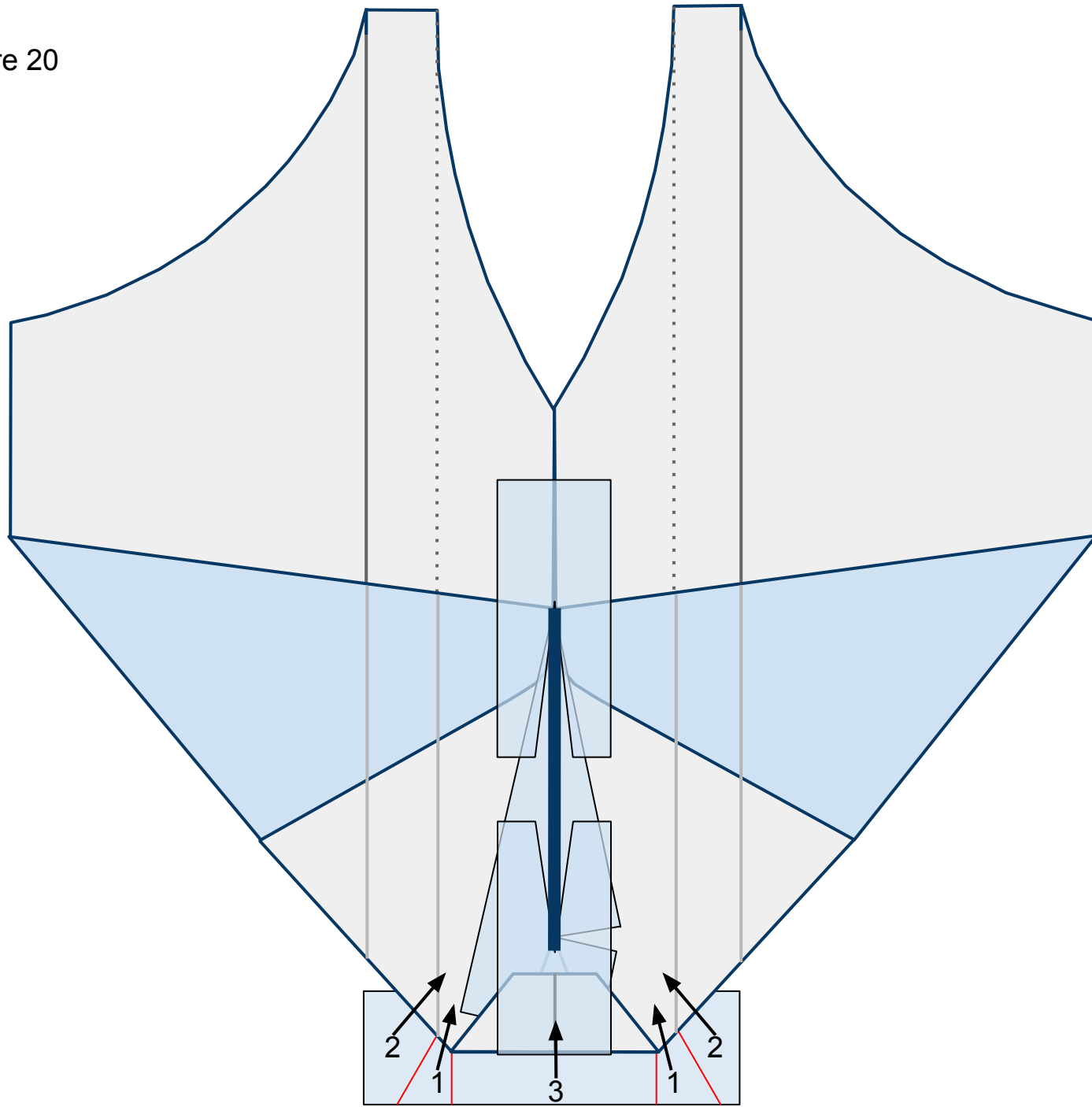
Figure 19



20

Flip the plane over so you can see the top side. Fold the flaps of tape, in number order, up and in as shown in Figure 20 below. They will overlap each other once folded.

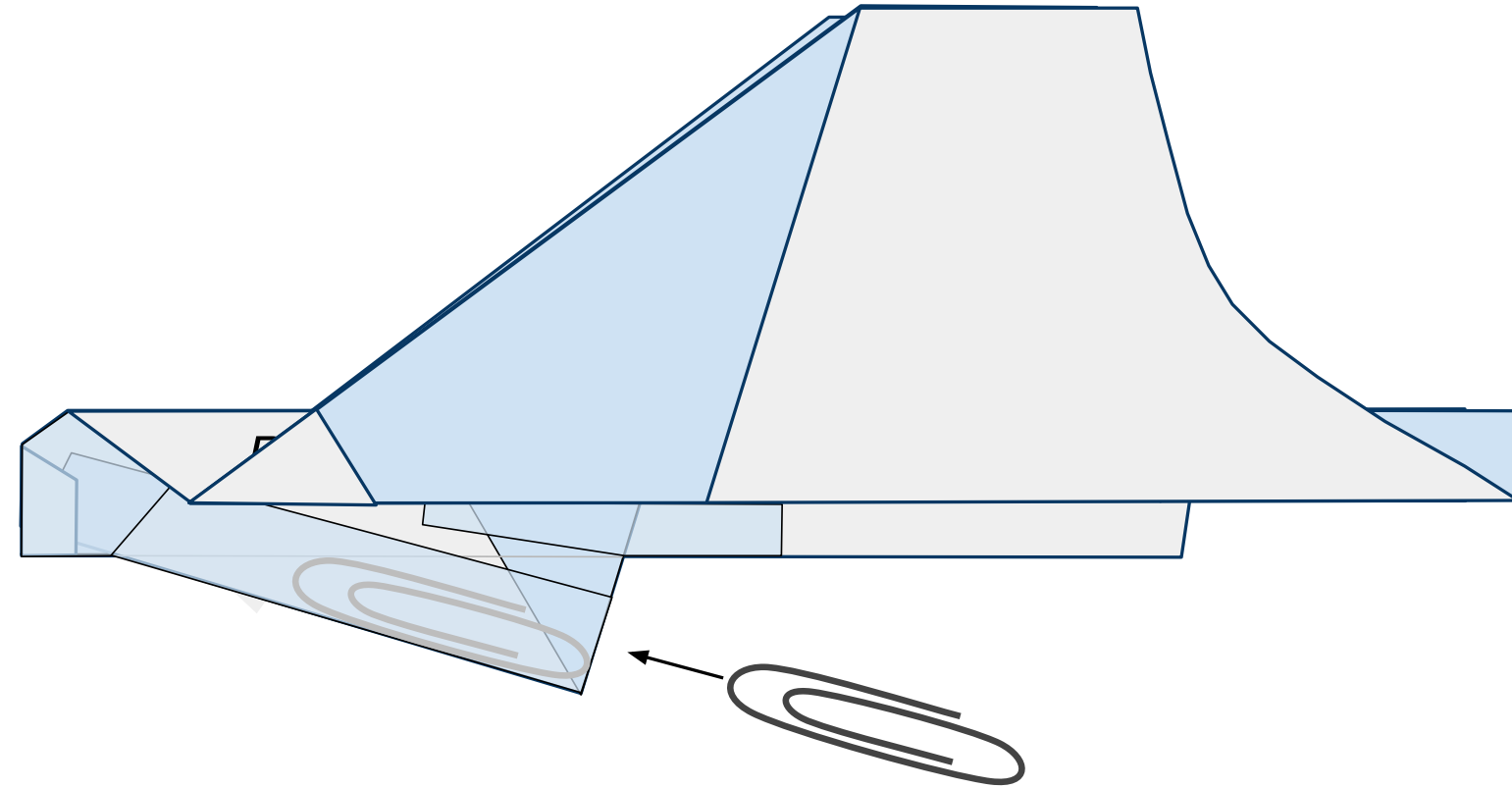
Figure 20



21

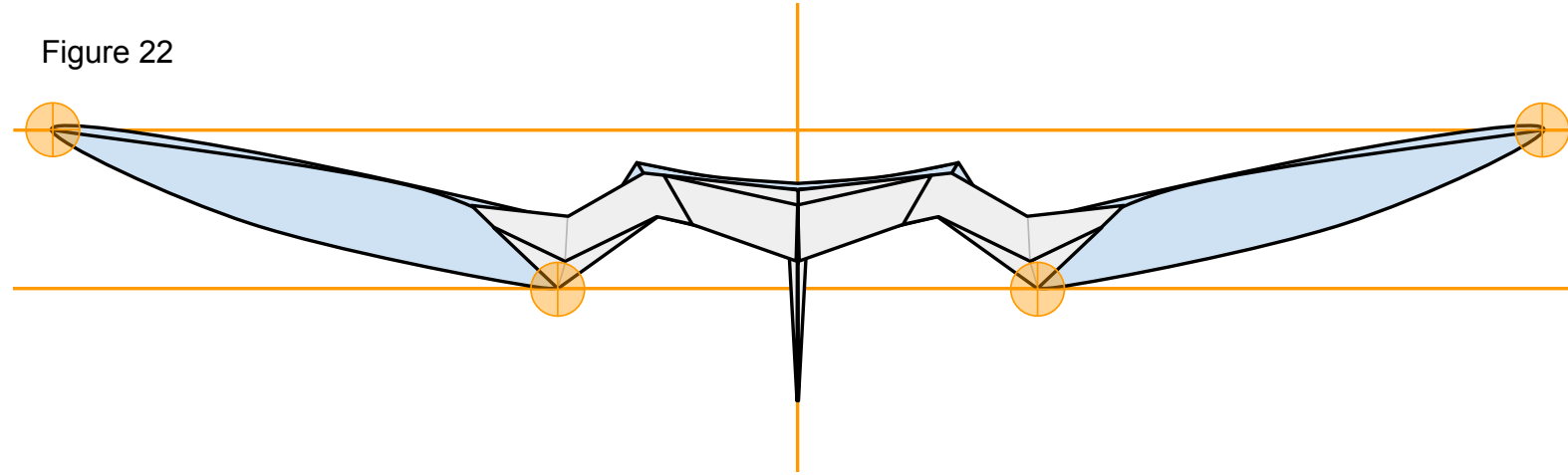
Turn the nose to the left and refold the wings keeping the tail up as shown. Insert a #1 size paper clip ***just*** inside the keel through the opening behind it as shown in Figure 21 below. Then seal the opening with a piece of tape.

Figure 21



Now that you have constructed the plane ensure that it has the proper shape. Take a look at your plane head-on from the nose. The plane should be symmetrical. Take a look at Figure 22 below. Note the **points of interest**. Note the curved shape of the paper on the underside of the wings. To achieve this, carefully stick one side of a pair of scissors up into the pocket on the underside of the wing from behind and gently mold the paper into the proper shape. Don't crease it.

Figure 22



We are almost ready to launch! I recommend taking a quick look at the **“PREFLIGHT”** section to get some good tips. All paper planes fly erratically on their first flights. They require refining to make them great. To do this, take a look at the **“FINE-TUNING”** section.